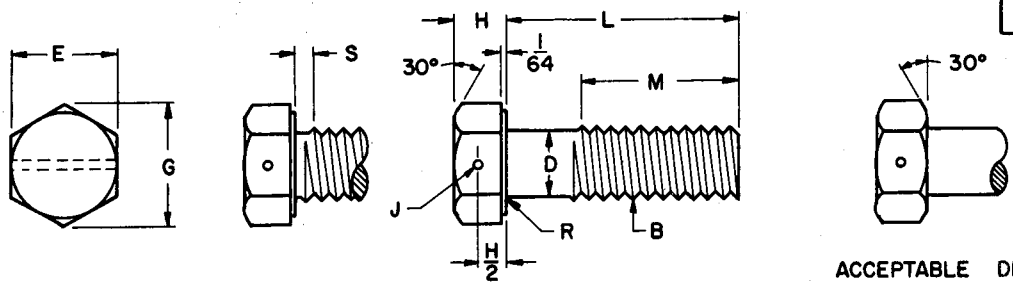
FED. SUP CLASS
5305



ACCEPTABLE DESIGN

	NOMINAL SIZE		1/4	5/16	3/8	7/16	1/2
B	THREADS PER INCH		28UNF-2A	24UNF-2A	24UNF-2A	20UNF-2A	20UNF-2A
D	BODY DIAMETER	Max	.2500	.3125	.3750	.4375	.5000
		Min	.2450	.3065	.3690	.4305	.4930
E	WIDTH ACROSS FLATS	Max	.4375	.5000	.5625	.6250	.7500
		Min	.4280	.4890	.5510	.6120	.7360
G	WIDTH ACROSS CORNERS	Max	.505	.577	.650	.722	.866
		Min	.488	.557	.628	.698	.840
H	HEAD HEIGHT	Max	.163	.211	.243	.291	.323
		Min	.150	.195	.226	.272	.302
R	RADIUS	Max	.023	.023	.023	.023	.023
		Min	.009	.009	.009	.009	.009
J	DIAMETER OF HOLE	+.006 -.002	.0625	.0625	.0625	.0625	.094
M	THREAD LENGTH	Max	15/16	1-1/16	1-3/16	1-5/16	1-7/16
		Min	3/4	7/8	1	1-1/8	1-1/4
S	UNTHREADED LENGTH*	Max	1/8	5/32	5/32	3/16	3/16

L LENGTH	Tolerance	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number
3/8		1	27			
7/16		2	28			
1/2		3	29	55		
9/16	+0	4	30	56		
5/8	-1/32	5	31	57		
3/4		6	32	58	84	110
7/8		7	33	59	85	111
1		8	34	60	86	112
1-1/4		9	35	61	87	113
1-1/2	+0	10	36	62	88	114
1-3/4	-1/16	11	37	63	89	115
2		12	38	64	90	116
2-1/4		13	39	65	91	117
2-1/2		14	40	66	92	118
2-3/4		15	41	67	93	119
3		16	42	68	94	120
3-1/4		17	43	69	95	121
3-1/2		18	44	70	96	122
3-3/4	+0	19	45	71	97	123
4	-3/32	20	46	72	98	124
4-1/4		21	47	73	99	125
4-1/2		22	48	74	100	126
4-3/4		23	49	75	101	127
5		24	50	76	102	128
5-1/2						129
6						130

MATERIAL: Aluminum alloy, Specification QQ-A-268, heat treated in accordance with MIL-A-6088.
SURFACE TREATMENT: Anodized, Type I, in accordance with Specification MIL-A-8625.
THREADS: In accordance with Screw-Thread Standards for Federal Services Handbook H-28.
UNTHREADED LENGTH: *The unthreaded length of screws too short for the thread length shown, shall not exceed the maximum shown.
DIMENSIONS: All dimensions are in inches unless otherwise specified.
PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS51102-1.
NOTES: (1) Referenced documents shall be of the issue in effect on date of invitations for bid.
(2) This standard takes precedence over documents referenced herein.
This document has been promulgated by the Department of Defense as the Military Standard to limit the selection of the item, product, or design covered herein in engineering, design, and procurement. This standard shall become effective not later than 90 days after the latest date of approval shown.

Ⓐ CANCELLED AFTER 6 JUNE 63 NO REPLACEMENT

Ⓐ FOR CHANGES SEE SHEETS 1 AND 2.

APPROVED 16 JULY, 58 REVISED Ⓐ 6 JUNE 63

NOTICE—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A.	WC
Other Cust	SHIPS ASD

TITLE
SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), HEAD DRILLED FOR LOCKING WIRE, ALUMINUM ALLOY, ANODIZED, UNF-2A

MILITARY STANDARD
MS 51102

PROCUREMENT SPECIFICATION
FF-S-85

SUPERSEDES:

SHEET 1 OF 2

DD FORM 1 SEP 67 672-1 (Continued) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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FED. SUP CLASS
5305

	NOMINAL SIZE		5/8	3/4	7/8	1
B	THREADS PER INCH		18UNF-2A	16UNF-2A	14UNF-2A	12UNF-2A
D	BODY DIAMETER	Max	.6250	.7500	.8750	1.0000
		Min	.6170	.7410	.8660	.9900
E	WIDTH ACROSS FLATS	Max	.9375	1.1250	1.3125	1.5000
		Min	.9220	1.1000	1.2850	1.4690
G	WIDTH ACROSS CORNERS	Max	1.083	1.299	1.516	1.732
		Min	1.051	1.254	1.465	1.675
H	HEAD HEIGHT	Max	.403	.483	.563	.627
		Min	.378	.455	.531	.591
R	RADIUS	Max	.041	.041	.062	.093
		Min	.021	.021	.041	.062
J	DIAMETER OF HOLE	+.006 -.002	.094	.094	.125	.125
M	THREAD LENGTH	Max	1-23/32	2	2-9/32	2-9/16
		Min	1-1/2	1-3/4	2	2-1/4
S	UNTHREADED LENGTH*	Max	7/32	1/4	9/32	5/16

L LENGTH	Tolerance Size 3/4 and under	Tolerance Size 7/8 and over	Dash Number	Dash Number	Dash Number	Dash Number
9/16						
5/8	+0	+0				
3/4	-1/32	-1/16	133			
7/8			134			
1			135	157		
1-1/4			136	158	182	206
1-1/2	+0	+0	137	159	183	207
1-3/4	-1/16	-1/8	138	160	184	208
2			139	161	185	209
2-1/4			140	162	186	210
2-1/2			141	163	187	211
2-3/4			142	164	188	212
3			143	165	189	213
3-1/4			144	166	190	214
3-1/2			145	167	191	215
3-3/4	+0	+0	146	168	192	216
4	-3/32	-3/16	147	169	193	217
4-1/4			148	170	194	218
4-1/2			149	171	195	219
4-3/4			150	172	196	220
5			151	173	197	221
5-1/2			152	174	198	222
6			153	175	199	223

NOTE: For material, surface treatment, and other pertinent data, see sheet 1.

APPROVED 16 JULY 58 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 AND 2

Ⓐ CANCELLED AFTER 6 JUNE 63 NO REPLACEMENT

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P.A. WC
Other Cust SHIPS ASD

TITLE
SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), HEAD DRILLED FOR LOCKING WIRE, ALUMINUM ALLOY, ANODIZED, UNF-2A

MILITARY STANDARD
MS 51102

PROCUREMENT SPECIFICATION
FF-S-85

SUPERSEDES:

SHEET 2 OF

DD FORM 1 SEP 57 672-1 (Obsolete) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

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